

Minimum Material Parameter Set for Reliable VARI Filling Simulations

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Abstract

The Vacuum-Assisted Resin Infusion belongs to the family of Liquid Composite Molding processes. Dry fibrous preforms are placed into a single-sided mold, which is then sealed with a flexible bag. The applied vacuum draws the resin through the fiber preform. A highly permeable flow distribution medium is commonly placed on top of the fibrous preform to facilitate resin flow. During mold design, filling simulations are used to study different placements of inlets and outlet to ensure complete filling and to estimate the filling time. A pheno-numerical simulation strategy is employed. Material parameters are either determined beforehand (porosities) or fitted from characterization experiments (in-plane and out-of-plane permeabilities) and then used as input parameters for the filling simulations of the actual part to be manufactured. For the current study, the component under investigation is a winglet rib with bounding box $510 \times 130 \times 60 \text{ mm}^3$ with wall thickness between 2 and 3 mm. The study shows that it is possible to extract all relevant material parameters for a filling simulation from a single characterization experiment when using a quasi-isotropic layup. Furthermore, generalizations for non-quasi-isotropic layups are proposed. All filling simulations are carried out using the software package LCMsim v3.

Introduction

Manufacturing simulations, such as those predicting process-induced deformations or the flow front propagation in Liquid Composite Molding (LCM), help improve part quality. A pheno-numerical simulation approach is often employed. Characterization experiments must be conducted beforehand to identify all required model parameters for the actual part simulation.

Vacuum-Assisted Resin Infusion (VARI) is a manufacturing method where a dry fibrous preform (FP) is placed into a mold which is subsequently sealed with a flexible bag by drawing vacuum. The vacuum then draws resin through the preform. A distribution medium (DM) partially or fully covers the FP to accelerate the filling process.

VARI with a DM is a complex process, and simulating the flow front propagation requires identifying multiple model parameters. The goal of this study is to reduce the number of re-

quired model parameters and to design an experiment that allows all relevant parameters to be determined by comparing experimental and simulated flow fronts at different time instances.

A winglet rib with bounding box $510 \times 130 \times 60 \text{ mm}^3$ and regions of varying wall thickness, as shown in the following figure, is considered here as a representative example of a VARI part.

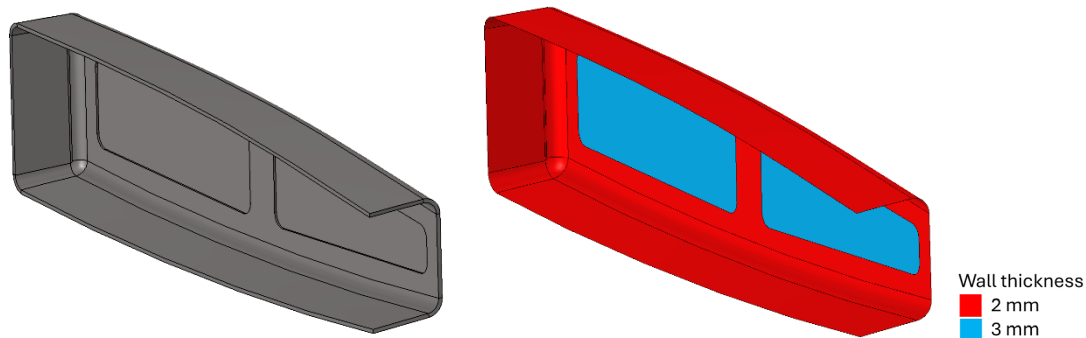


Fig. 1: Solid winglet geometry (left) to be manufactured with quasi-isotropic layup. Wall thickness distribution shown with the shell model (right).

A dry fibrous preform is placed in a quasi-isotropic layup into a female mold. For manufacturing with VARI, a DM covers the entire FP and the flanges of the mold. Inlet and outlets are positioned, and then the entire manufacturing assembly is sealed with a flexible vacuum. A schematic of the manufacturing concept is shown in the following figure.

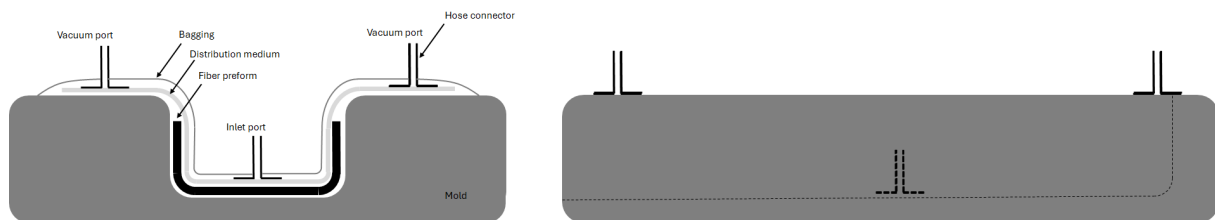


Fig. 2: Female mold concept for winglet rib with central inlet and outlets at the corners. Side view showing the details (left) and front view showing the distribution of central inlet and outlets at the corners (right).

To investigate different manufacturing concepts (i.e. various inlet and outlet positions) and accurately estimate the filling time, the FP and DM parameters (porosity and permeability) must be determined in advance through a characterization experiment.

The simulation model is used first to fit parameters by comparing the results of the characterization experiment with the corresponding simulation, and later to perform filling simulations during mold design.

The requirements shown in the following table were defined for both the characterization experiment and the simulation model.

Table 1: Requirements for a characterization experiment and the simulation model

Requirements for the characterization experiment	Requirements for the simulation model
• One experiment must reveal all unknown material parameters • Easy to carry out • Minimize the number of sensors, flow front position is camera recorded and flow front positions seen from top and bottom are extracted at different time instances • Fluid properties (mass density, dynamic viscosity) of the test fluid must be known	• Minimize the number of unknown material parameters • Small simulation time (because multiple runs required for parameter fitting), simple workflow and free software

Since a single experiment must reveal all unknown material parameters, the characterization setup must include one region with DM and one region without DM. To simplify the process, a linear flow experiment is preferred over a radial flow experiment (although two characterization experiments for non-quasi-isotropic layup). The mold should be made of a glass to ensure visibility of the flow front at both the top and bottom surfaces. A pressure sensor placed in the catch pot is used to measure the degree of vacuum and two distance measurement sensors (such as those based on the linear variable differential transformer (LVDT) technology) are required (one in the region with DM and one in the region without DM).

The number of unknown material parameters can be reduced by combining all consumable layers (e.g. flow distribution medium and peel ply) to the DM and assuming isotropic permeability for the DM. Furthermore, the exponent in the Cozeny-Karman equation for FP permeability is set equal to $n=2$, and the pressure-dependent porosity change is described only by the porosity at maximum vacuum and at ambient pressure. Free software packages for filling simulations of this type include the general purpose CFD package OpenFOAM [1] and the specialized filling simulation tool LCMsim [2]. The latter, in its version v3 [3], was selected due to its simpler workflow for filling simulations.

Materials and Methods

The following figure illustrates the concept for a characterization experiment. In the left section, a fast horizontal flow occurs through the DM while the FP is filled in vertical direction. Consequently, the flow front visible from below lags behind the one on the top side. Once the DM is completely saturated, this so-called lead-lag effect diminishes, and after a certain time, the flow front propagates from left to right through the FP in the right section.

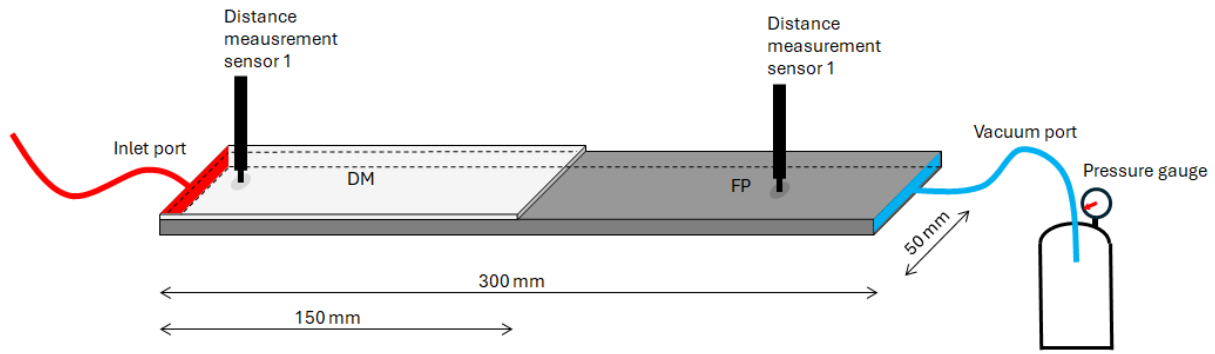


Fig. 3: Test case setup and dimensions.

The distance measurement sensor on the DM should be positioned as close as possible to the inlet to ensure that ambient pressure is reached there at the end of the experiment, when the cavity is completely filled. At this stage the pressure and porosity at ambient pressure (denoted by index 1) are extracted. This also implies that hose lengths should be kept short to minimize pressure losses.

To validate the proposed approach, published experimental data was used. Case 2 from reference [4] describes such an experiment. All data required to be extracted from a characterization experiment is available there and is presented in the following tables (thickness and porosity values, as well as flow front position observed from the top and bottom). Fluid parameters used are mass density $\rho=920 \text{ kg/m}^3$ and dynamic viscosity $\mu=0.3 \text{ Pa s}$.

Table 2: Extracted thickness and porosity data from experiment

Thickness of FP at ambient pressure (100000 Pa)	$t_{FP,1}$	2.2 mm
Thickness of FP at 23000 Pa	$t_{FP,0}$	3.2 mm
Thickness of DM	t_{DM}	0.55 mm
Porosity of FP at ambient pressure (100000 Pa)	$\epsilon_{FP,1}$	0.41
Porosity of FP at 23000 Pa	$\epsilon_{FP,0}$	0.46
Porosity of DM	ϵ_{DM}	0.5

Table 3: Extracted flow front position at different time instances seen from top (DM) and bottom (FP).

Time [s]	Flow front at top of DM [mm]	Flow front at bottom of FP [mm]
9 s	90	0
23 s	128	37
32 s	146	56
44 s	150	77
135 s	174	174
230 s	198	198

The porosity values were calculated based on the known areal weight of the plies, number of the plies, the fiber material mass density, and the total thickness according to $\varepsilon = 1 - (n A_f)/(\rho_f t)$. For a combined stack of consumables, the porosity is taken as the thickness-averaged porosity of the two consumables, provided guessed thickness values for the individual consumables are known or estimated. The two (pressure, porosity) tuples are fitted to a non-linear curve as illustrated in the following figure.

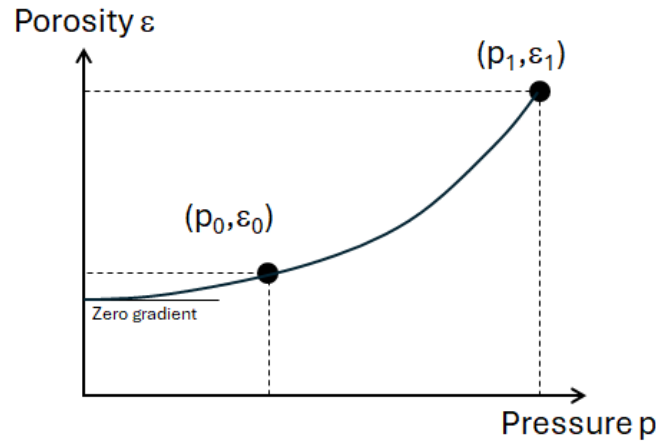


Fig. 4: Construction of a non-linear $\varepsilon(p)$ curve with zero gradient at vacuum pressure (left).

Results and Discussion

The extracted thickness, porosity and flow front position data from the referenced paper correspond to data obtained from a characterization experiment. A good parameter fit for

- the isotropic permeability K_{DM} of the DM,
- the parameter $k_{QI,FP}$ in the Cozeny-Karman equation for the quasi-isotropic horizontal

$$\text{permeability } K_{QI,FP} = k_{QI,FP} \frac{(\varepsilon_{FP})^3}{(1-\varepsilon_{FP})^2}$$

- the parameter $k_{Z,FP}$ in the Cozeny-Karman equation for the vertical permeability

$$K_{Z,FP} = k_{Z,FP} \frac{(\varepsilon_{FP})^3}{(1-\varepsilon_{FP})^2}$$

yields the material parameters: $K_{DM}=14 \cdot 10^{-10} \text{ m}^2$, $k_{QI,FP}=8 \cdot 10^{-11} \text{ m}^2$, $k_{Z,FP}=1.25 \cdot 10^{-12} \text{ m}^2$. This material parameter set produces a flow front propagation as shown in the following figure.

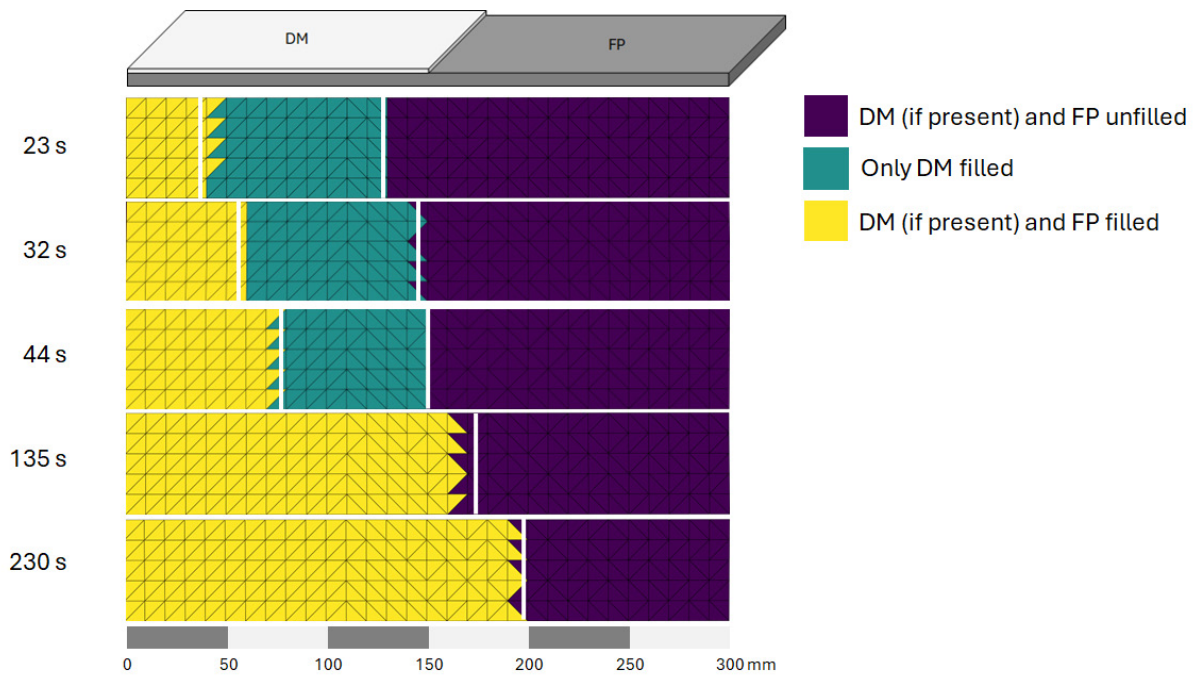


Fig. 5: Flow front for the fitted material parameters. The white bars indicate the flow front from the experiments.

In the original publication, similar parameters were reported: $k_{QI,FP}=8.14 \cdot 10^{-11} \text{ m}^2$ and $k_{Z,FP}=2.49 \cdot 10^{-12} \text{ m}^2$ for the FP. In contrast to the presented approach, the DM properties in that study differ between the horizontal and vertical directions with $K_{DM,QI}=9.5 \cdot 10^{-10} \text{ m}^2$ and $K_{DM,Z}=1 \cdot 10^{-13} \text{ m}^2$. The FP properties in that work were derived from a comprehensive material characterization program involving compaction experiments, while the DM properties were adjusted to ensure that experimental and simulated flow fronts matched. Considering this, the fitted material parameters obtained here show very good agreement with the published values in the original publication.

Due to the design of the experiment, there is always only a single solution triple $(K_{DM}, k_{QI,FP}, k_{Z,FP})$, or rather a box-shaped interval around it, within which all parameter combinations provide a good fit to the flow front propagation.

The following picture illustrates how a potential test stand could look. Four distance measurement devices are arranged equidistantly (left). The sensors are zeroed using an empty plate. Afterwards, the material stack is prepared on the plate (middle) and placed back on the test rig. When both top and bottom flow fronts can be tracked from above (right) one camera is sufficient. Otherwise, a glass plate must be used as mold and a second camera is required.

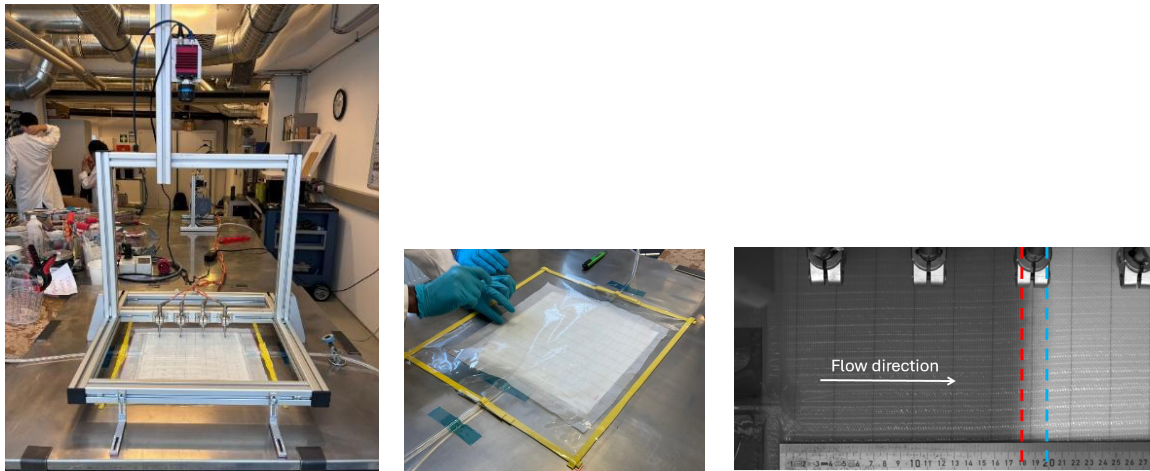


Fig. 6: Full test stand (left), material preparation (middle) and flow front propagation (right). The full saturated front (red dashed line) and the flow front in the DM (blue dashed line) are indicated.

If the FP layup is not quasi-isotropic, two separate characterization experiments must be performed, one in each principal flow direction.

Conclusion

Pheno-numerical simulations are the predominant method used for LCM filling simulations. Consequently, model parameters must be determined through characterization experiments prior to simulating the actual part. This study demonstrates how the number of model parameters required for VARI filling simulations when the FP is partially or fully covered by a DM can be reduced and characterized using only a single experiment. If the layup is not quasi-isotropic or consists of regions with different layups or thickness, additional experiments are required accordingly.

References

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